

Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \dots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \dots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period.

SHC by [Chen et al. \[2024\]](#) constructs

$$N = T_0 - m - n + 1$$

overlapping temporal segments of length m . Each segment $\tilde{\mathbf{y}}_{[i, i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix

$$\tilde{\mathbf{Y}}_{\text{pre}} := [\tilde{\mathbf{y}}_{[1, m]} \quad \tilde{\mathbf{y}}_{[2, m+1]} \quad \dots \quad \tilde{\mathbf{y}}_{[N, N+m-1]}] \in \mathbb{R}^{m \times N}.$$

Example 0.1. Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:

$$N = 60 - 12 - 6 + 1 = 43.$$

Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\text{argmin}} \quad & \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC solution, and $\mathbf{w}^{\text{ASHC}}$ the ASHC solution:

$$\mathbf{w}^{\text{ASHC}} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \left\| \tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w} \right\|^2 + \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\} \quad \text{s.t.} \quad \sum_{i=1}^N w_i = 1,$$

where $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is the penalty parameter for the low-variance directions, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of the Gram matrix $\mathbf{G} = \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

Assumptions

Assumption 0.1 (Linearity with noise). *There exists $\mathbf{w}^* \in \mathbb{R}^N$ such that*

$$\tilde{\mathbf{y}} = \tilde{\mathbf{Y}}\mathbf{w}^* + \boldsymbol{\eta},$$

where $\boldsymbol{\eta} \in \mathbb{R}^m$ is a mean-zero noise vector.

Assumption 0.2 (Bounded noise). $\|\boldsymbol{\eta}\| \leq \delta$ for some $\delta > 0$.

Assumption 0.3 (Smoothness of latent trend). *Let $\ell(t)$ denote the latent smooth component of the treated potential outcome. Suppose $\ell(t)$ is $H + 1$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t and some $H \geq 3$.*

Assumption 0.4 (Operator norm bound). $\|\tilde{\mathbf{Y}}\| \leq \gamma$.

Assumption 0.5 (Simplex feasibility). $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1, w_i \geq 0\}$.

Lemma 0.1 (Deviation from SHC). *Let $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$. Then*

$$\|\Delta\| \leq \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \epsilon_{SHC}.$$

where $\epsilon_{SHC} := \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ and $\|\mathbf{C}_0\|^2$ is the operator norm of $\mathbf{C}_0^\top \mathbf{C}_0$.

Proof. To analyze the ASHC optimization problem, recall the objective function:

$$J(\mathbf{w}) = \frac{1}{2\lambda} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{SHC}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|^2, \quad (0.1)$$

subject to the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$.

To enforce this constraint, we form the Lagrangian:

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1), \quad (0.2)$$

where μ is a Lagrange multiplier ensuring feasibility of $\mathbf{w}$.

We compute the gradient of $\mathcal{L}$ with respect to $\mathbf{w}$ by differentiating each term separately.

For the first term, define the residual vector:

$$r(\mathbf{w}) = \tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}. \quad (0.3)$$

The fit term is then:

$$\frac{1}{2\lambda} \|r(\mathbf{w})\|^2 = \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w}). \quad (0.4)$$

Using the chain rule, the gradient with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}} \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w}) = \frac{1}{\lambda} \nabla_{\mathbf{w}} r(\mathbf{w})^\top r(\mathbf{w}). \quad (0.5)$$

Since

$$r(\mathbf{w}) = \tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}, \quad (0.6)$$

we have

$$\nabla_{\mathbf{w}} r(\mathbf{w}) = -\tilde{\mathbf{Y}}. \quad (0.7)$$

Therefore,

$$\nabla_{\mathbf{w}} \frac{1}{2\lambda} \|r(\mathbf{w})\|^2 = -\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top r(\mathbf{w}) = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{Y}}\mathbf{w} - \tilde{\mathbf{y}}). \quad (0.8)$$

Next, consider the deviation penalty term:

$$\|\mathbf{w} - \mathbf{w}^{SHC}\|^2 = (\mathbf{w} - \mathbf{w}^{SHC})^\top (\mathbf{w} - \mathbf{w}^{SHC}). \quad (0.9)$$

Its gradient with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\text{SHC}}). \quad (0.10)$$

Notice that the Hessian of this term is $2\mathbf{I}$, because differentiating again with respect to $\mathbf{w}$ gives:

$$\nabla_{\mathbf{w}}^2 \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 = 2\mathbf{I}. \quad (0.11)$$

Here $\mathbf{I}$ is the identity matrix, representing curvature equally in all coordinate directions. Thus the $2\mathbf{I}$ in M directly reflects the Hessian of this quadratic deviation penalty.

For the Gram penalty term, rewrite it as:

$$\varsigma \|\mathbf{C}_0^\top \mathbf{w}\|^2 = \varsigma \mathbf{w}^\top \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}. \quad (0.12)$$

Since $\mathbf{C}_0 \mathbf{C}_0^\top$ is symmetric, its gradient is:

$$\nabla_{\mathbf{w}} \varsigma \mathbf{w}^\top \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w} = 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}. \quad (0.13)$$

Finally, the affine constraint term differentiates as:

$$\nabla_{\mathbf{w}} \mu (\mathbf{1}^\top \mathbf{w} - 1) = \mu \mathbf{1}. \quad (0.14)$$

Collecting all these gradient terms, the stationarity (first-order optimality) condition requires:

$$\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{Y}} \mathbf{w} - \tilde{\mathbf{y}}) + 2(\mathbf{w} - \mathbf{w}^{\text{SHC}}) + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w} + \mu \mathbf{1} = \mathbf{0}. \quad (0.15)$$

Rearranging terms, we have:

$$\left(\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \right) \mathbf{w} = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{y}} + 2\mathbf{w}^{\text{SHC}} - \mu \mathbf{1}. \quad (0.16)$$

Define the matrix

$$M = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top, \quad (0.17)$$

which summarizes the curvature of the Lagrangian. It corresponds to the Hessian of the objective function $J(\mathbf{w})$ plus the identity-weighted contribution of the deviation penalty.

The solution $\mathbf{w}^{\text{ASHC}}$ satisfies the first-order condition:

$$M \mathbf{w}^{\text{ASHC}} = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{y}} + 2\mathbf{w}^{\text{SHC}} - \mu \mathbf{1}. \quad (0.18)$$

Evaluating the same matrix M at $\mathbf{w}^{\text{SHC}}$, which is not necessarily the ASHC solution, gives:

$$M \mathbf{w}^{\text{SHC}} = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} \mathbf{w}^{\text{SHC}} + 2\mathbf{w}^{\text{SHC}} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}^{\text{SHC}}. \quad (0.19)$$

Subtracting this equation from the previous one, we get

$$M \mathbf{w}^{\text{ASHC}} - M \mathbf{w}^{\text{SHC}} = \left(\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{y}} + 2\mathbf{w}^{\text{SHC}} - \mu \mathbf{1} \right) - \left(\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} \mathbf{w}^{\text{SHC}} + 2\mathbf{w}^{\text{SHC}} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}^{\text{SHC}} \right). \quad (0.20)$$

Simplifying the right-hand side by canceling the $2\mathbf{w}^{\text{SHC}}$ terms, we obtain

$$M(\mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}}) = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w}^{\text{SHC}}) - \mu \mathbf{1}. \quad (0.21)$$

Define the deviation

$$\Delta = \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}}, \quad (0.22)$$

and the residual vector

$$r = \tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}. \quad (0.23)$$

Since both $\mathbf{w}^{\text{ASHC}}$ and $\mathbf{w}^{\text{SHC}}$ satisfy the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$, their difference Δ lies in the hyperplane orthogonal to $\mathbf{1}$:

$$\mathbf{1}^\top \Delta = 0. \quad (0.24)$$

Taking the inner product of both sides of the previous equation with Δ gives

$$\Delta^\top M \Delta = \frac{1}{\lambda} \Delta^\top \tilde{\mathbf{Y}}^\top r - \mu \underbrace{\Delta^\top \mathbf{1}}_{=0} = \frac{1}{\lambda} \Delta^\top \tilde{\mathbf{Y}}^\top r. \quad (0.25)$$

Here, the right-hand side is an inner product between the deviation vector Δ and the vector $\tilde{\mathbf{Y}}^\top r$. Because inner products can be difficult to control directly, we apply the Cauchy–Schwarz inequality to bound it:

$$|\Delta^\top \tilde{\mathbf{Y}}^\top r| \leq \|\Delta\| \|\tilde{\mathbf{Y}}^\top r\|. \quad (0.26)$$

Thus,

$$\Delta^\top M \Delta \leq \frac{1}{\lambda} \|\Delta\| \|\tilde{\mathbf{Y}}^\top r\|. \quad (0.27)$$

By assumption, the operator norm of $\tilde{\mathbf{Y}}$ is bounded by γ , and the residual norm is $\epsilon_{\text{SHC}} = \|r\|$, so

$$\|\tilde{\mathbf{Y}}^\top r\| \leq \|\tilde{\mathbf{Y}}\| \|r\| \leq \gamma \epsilon_{\text{SHC}}. \quad (0.28)$$

Thus,

$$\Delta^\top M \Delta \leq \frac{\gamma}{\lambda} \|\Delta\| \epsilon_{\text{SHC}}. \quad (0.29)$$

On the other hand, since M is positive definite, its smallest eigenvalue $\lambda_{\min}(M)$ guarantees

$$\Delta^\top M \Delta \geq \lambda_{\min}(M) \|\Delta\|^2. \quad (0.30)$$

Putting these two inequalities together, we sandwich the quadratic form:

$$\lambda_{\min}(M) \|\Delta\|^2 \leq \Delta^\top M \Delta \leq \frac{\gamma}{\lambda} \|\Delta\| \epsilon_{\text{SHC}}. \quad (0.31)$$

Dividing by $\|\Delta\| > 0$ (the trivial case $\Delta = 0$ is clear) gives the bound

$$\|\Delta\| \leq \frac{\gamma}{\lambda \lambda_{\min}(M)} \epsilon_{\text{SHC}}. \quad (0.32)$$

Recall that

$$M = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top. \quad (0.33)$$

Its smallest eigenvalue is at least

$$\lambda_{\min}(M) \geq 2 + 2\varsigma \lambda_{\min}(\mathbf{C}_0 \mathbf{C}_0^\top) \geq 2 + 2\varsigma \|\mathbf{C}_0\|^2, \quad (0.34)$$

where $\|\mathbf{C}_0\|^2$ denotes the operator norm squared.

Therefore, the final bound on the deviation is

$$\|\Delta\| \leq \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \epsilon_{\text{SHC}}. \quad (0.35)$$

This completes the proof, showing that the difference between the ASHC and SHC weights is controlled by the residual fit error of SHC, scaled by a factor depending on the tuning parameters λ and ς . $\square$

Remark 0.1. *This result shows that the difference between the ASHC weights and the original SHC weights — the deviation Δ — is directly controlled by how well the SHC fit performs (measured by the residual error ϵ_{SHC}) and the strength of the regularization penalties (λ and ς).*

Importantly, the affine constraint (weights summing to one) is enforced by the Lagrange multiplier μ , which acts like a balancing force that keeps the weights feasible but does not increase the deviation size.

Thus, the ASHC estimator remains close to the SHC baseline, with the regularization terms effectively shrinking the deviation to be small whenever the SHC fit is good. This gives applied researchers a clear sense that ASHC improves stability without drastically changing the solution unless necessary, ensuring robust predictions.

Lemma 0.2. *Under (A4), the error from deviating off SHC weights satisfies*

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|.$$

Proof. By definition, $\Delta = \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}} \in \mathbb{R}^N$. From Assumption (A4), the operator (spectral) norm of the donor matrix satisfies

$$\|\tilde{\mathbf{Y}}\| \leq \gamma,$$

where the operator norm is

$$\|\tilde{\mathbf{Y}}\| = \sup_{\mathbf{v} \neq 0} \frac{\|\tilde{\mathbf{Y}}\mathbf{v}\|}{\|\mathbf{v}\|}.$$

Applying this inequality to $\mathbf{v} = \Delta$ gives

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \|\tilde{\mathbf{Y}}\| \|\Delta\| \leq \gamma\|\Delta\|.$$

This establishes the claim. $\square$

Remark 0.2. *Lemma 0.2 ensures that deviations of the ASHC weights from the SHC weights, measured in ℓ_2 norm, translate into controlled deviations in the predicted outcomes. The operator norm bound (A4) prevents $\tilde{\mathbf{Y}}$ from amplifying small changes in the weights disproportionately.*

Lemma 0.3. *Under (A3), for horizon $k > 0$,*

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq b_\epsilon(H, k) := \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof. We show that the SHC approximation error in predicting the smooth latent trend $\ell(\cdot)$ is of order $|k|^{H+1}$. The argument proceeds in four steps.

Step 1. Taylor expansion of the latent trend. Let $\ell(t)$ denote the latent smooth component of the treated outcome. By Assumption (A3), $\ell(t)$ is $(H+1)$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t . Expanding ℓ around T_0 using Taylor's theorem gives

$$\ell(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

where the remainder term is

$$R_{H+1}(k) = \frac{\ell^{(H+1)}(\xi)}{(H+1)!} k^{H+1}, \quad \text{for some } \xi \in [T_0, T_0 + k].$$

Step 2. Bound the Taylor remainder. Since $|\ell^{(H+1)}(t)| \leq \epsilon$ for all t , the remainder is bounded by

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Step 3. Representation of the SHC approximation. The SHC predictor $\ell^{\text{SHC}}(T_0 + k)$ is constructed as a convex combination of donor segments. By design, these donor segments come from smoothed pre-treatment trajectories that share the same latent trend structure $\ell(\cdot)$. Thus, when we form ℓ^{SHC} , the lower-order terms of the Taylor expansion (up to order H) are matched exactly in expectation, and the residual discrepancy is of the same order as the Taylor remainder.

Formally, we may write

$$\ell^{\text{SHC}}(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R'_{H+1}(k),$$

where $R'_{H+1}(k)$ is another error term of order $|k|^{H+1}$.

Step 4. Combine the two remainder terms. Subtracting the SHC expansion from the true expansion gives

$$\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k) = R_{H+1}(k) - R'_{H+1}(k).$$

Applying the triangle inequality,

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq |R_{H+1}(k)| + |R'_{H+1}(k)|.$$

Each remainder is bounded by $\frac{\epsilon}{(H+1)!} |k|^{H+1}$, so together we obtain

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1}.$$

This completes the proof. □

Remark 0.3. Lemma 0.3 formalizes the idea that the SHC predictor inherits the smoothness of the latent trend $\ell(\cdot)$. Because SHC is built from overlapping donor segments that reflect the same underlying smooth dynamics, the approximation reproduces the first H terms of the Taylor expansion around T_0 . The only discrepancy comes from the $(H+1)$ -st order remainder, which is small when the latent trend is sufficiently smooth. Thus, the difference between ℓ and ℓ^{SHC} grows only polynomially in the forecast horizon k , and the error is of order $|k|^{H+1}$ rather than linear or quadratic in k . This ensures that SHC extrapolates the smooth component accurately over short to moderate horizons.

Main Result

Proposition 0.1. Under assumptions (A1)–(A5), the ASHC prediction error is bounded by

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \right) + b_\epsilon(H, k) + \delta.$$

Proof. We want to establish an upper bound on the prediction error of the ASHC estimator,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\|.$$

The argument proceeds in four steps.

Step 1. Break down the error into two components. Let $\mathbf{w}^{\text{SHC}}$ denote the SHC weights and define

$$\Delta := \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}}.$$

By the triangle inequality,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \underbrace{\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{SHC}}\|}_{\epsilon_{\text{SHC}}} + \|\tilde{\mathbf{Y}}\Delta\|.$$

Step 2. Bound the contribution of $\|\tilde{\mathbf{Y}}\Delta\|$. By Lemma 0.2, we know that

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|.$$

Step 3. Bound the deviation $\|\Delta\|$. By Lemma 0.1, the deviation satisfies

$$\|\Delta\| \leq \frac{\epsilon_{\text{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Step 4. Account for smoothness bias and noise. From Lemma 0.3, the smoothness bias is bounded by

$$b_\epsilon(H, k) = \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

From Assumption (A2), the noise contributes at most

$$\delta.$$

Step 5. Putting the pieces together. Substituting the bound from Step 2 into Step 1 gives:

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} + \gamma\|\Delta\|.$$

Now substitute the bound for $\|\Delta\|$ from Step 3:

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} + \gamma \cdot \frac{\epsilon_{\text{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Next, add the smoothness bias and noise bounds from Step 4:

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} + \gamma \cdot \frac{\epsilon_{\text{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} + b_\epsilon(H, k) + \delta.$$

Step 6. Simplify the bound. Factor out ϵ_{SHC} from the first two terms:

$$\epsilon_{\text{SHC}} + \gamma \cdot \frac{\epsilon_{\text{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} = \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \right).$$

Therefore,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} \right) + b_\epsilon(H, k) + \delta.$$

This completes the proof. $\square$

Remark 0.4. The proof explicitly shows how each component contributes to the final bound: (1) the SHC baseline error, (2) the ASHC deviation penalty controlled by λ and ς , (3) the smoothness bias from Taylor approximation, and (4) the bounded noise. Factoring ϵ_{SHC} at the end highlights how ASHC scales the SHC error by at most a small multiplicative factor depending on the regularization strength.

Asymptotic Results

Theorem 0.1. Under (A1)–(A5), as $m \rightarrow \infty$ and $H \rightarrow \infty$, for fixed $k > 0$,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \rightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 0.1, we have

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta.$$

Step 1. Behavior of ϵ_{SHC} . By (A3), $\ell(t)$ is $(H+1)$ -times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\text{SHC}} \rightarrow 0 \quad \text{as } m \rightarrow \infty, H \rightarrow \infty.$$

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Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\text{SHC}} \rightarrow 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \epsilon_{\text{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H, k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Thus

$$b_\epsilon(H, k) \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1} \rightarrow 0 \quad \text{as } H \rightarrow \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \leq \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2 , then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m \rightarrow \infty} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \leq \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. $\square$

Corollary 0.1. Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ . If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \rightarrow \infty$,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| = O_p\left(m^{-H/(2H+3)}\right).$$

For large H , this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. Part 1: Consistency. From Theorem 0.1, the error converges to δ , or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 0.1, we have

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta.$$

Step 1. Bias term. From Lemma 0.3,

$$b_\epsilon(H, k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p((mh_m)^{-1/2}).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\text{SHC}} = O_p(h_m^H + (mh_m)^{-1/2}).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\text{SHC}} = O_p(m^{-H/(2H+3)}).$$

Thus the total error is

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| = O_p(m^{-H/(2H+3)}).$$

Step 6. Near-parametric rate. As $H \rightarrow \infty$,

$$\frac{H}{2H+3} \rightarrow \frac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. □

Remark 0.5. The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \leq \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.

References

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